

## SAFETY DATA SHEET

Creation Date 22-September-2009

Revision Date 13-October-2023

Revision Number 3

### 1. Identification

**Product Name** Sodium hydroxide

**Cat No. :** SS264-1; XXSS2644LI; NC1802644

**Synonyms** Caustic soda; Lye.

**Recommended Use** Laboratory chemicals.

**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

**Importer/Distributor**  
Fisher Scientific  
112 Colonnade Road,  
Ottawa, ON K2E 7L6,  
Canada  
Tel: 1-800-234-7437

##### **Manufacturer**

Fisher Scientific Company  
One Reagent Lane  
Fair Lawn, NJ 07410  
Tel: (201) 796-7100

##### **Emergency Telephone Number**

CHEMTREC®, Inside the USA: 800-424-9300  
CHEMTREC®, Outside the USA: 001-703-527-3887

### 2. Hazard(s) identification

#### Classification

**WHMIS 2015 Classification** Classified as hazardous under the Hazardous Products Regulations (SOR/2015-17)

|                                                         |              |
|---------------------------------------------------------|--------------|
| <b>Corrosive to metals</b>                              | Category 1   |
| <b>Skin Corrosion/Irritation</b>                        | Category 1 A |
| <b>Serious Eye Damage/Eye Irritation</b>                | Category 1   |
| <b>Specific target organ toxicity (single exposure)</b> | Category 3   |
| Target Organs - Respiratory system.                     |              |

#### Label Elements

##### **Signal Word**

Danger

##### **Hazard Statements**

May be corrosive to metals  
Causes severe skin burns and eye damage  
May cause respiratory irritation

**Precautionary Statements****Prevention**

Do not breathe dust/fumes/gas/mist/vapours/spray  
Wash face, hands and any exposed skin thoroughly after handling  
Use only outdoors or in a well-ventilated area  
Wear protective gloves/protective clothing/eye protection/face protection

**Response**

IF SWALLOWED: Rinse mouth. Do NOT induce vomiting  
IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/ shower  
IF INHALED: Remove person to fresh air and keep comfortable for breathing  
IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
Immediately call a POISON CENTER/doctor  
Wash contaminated clothing before reuse

**Storage**

Store locked up  
Store in a well-ventilated place. Keep container tightly closed

**Disposal**

Dispose of contents/container to an approved waste disposal plant

### 3. Composition/Information on Ingredients

| Component        | CAS-No    | Weight % |
|------------------|-----------|----------|
| Water            | 7732-18-5 | 75 - 85  |
| Sodium hydroxide | 1310-73-2 | 15 - 25  |

### 4. First-aid measures

|                                        |                                                                                                                                                                                                                                                                                                   |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b>                  | Immediate medical attention is required. Show this safety data sheet to the doctor in attendance.                                                                                                                                                                                                 |
| <b>Eye Contact</b>                     | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Immediate medical attention is required.                                                                                                                                                                 |
| <b>Skin Contact</b>                    | Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.                                                                                                                                                                                       |
| <b>Inhalation</b>                      | Remove to fresh air. If breathing is difficult, give oxygen. Get medical attention.                                                                                                                                                                                                               |
| <b>Ingestion</b>                       | Do NOT induce vomiting. Get medical attention.                                                                                                                                                                                                                                                    |
| <b>Most important symptoms/effects</b> | Causes burns by all exposure routes. . Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation |
| <b>Notes to Physician</b>              | Treat symptomatically                                                                                                                                                                                                                                                                             |

### 5. Fire-fighting measures

**Suitable Extinguishing Media** CO<sub>2</sub>, dry chemical, dry sand, alcohol-resistant foam.

**Unsuitable Extinguishing Media** No information available

**Flash Point** Not applicable  
**Method -** No information available

**Autoignition Temperature** No information available

**Explosion Limits**

**Upper** No data available

**Lower** No data available

**Oxidizing Properties** Not oxidising

**Sensitivity to Mechanical Impact** No information available

**Sensitivity to Static Discharge** No information available

### Specific Hazards Arising from the Chemical

Thermal decomposition can lead to release of irritating gases and vapors. The product causes burns of eyes, skin and mucous membranes.

### Hazardous Combustion Products

Thermal decomposition can lead to release of irritating gases and vapors.

### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

### NFPA

**Health**  
3

**Flammability**  
0

**Instability**  
0

**Physical hazards**  
N/A

## 6. Accidental release measures

**Personal Precautions** Use personal protective equipment as required. Ensure adequate ventilation. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak.

**Environmental Precautions** Should not be released into the environment. See Section 12 for additional Ecological Information.

**Methods for Containment and Clean Up** Soak up with inert absorbent material. Keep in suitable, closed containers for disposal.

## 7. Handling and storage

**Handling** Use only under a chemical fume hood. Wear personal protective equipment/face protection. Do not breathe mist/vapors/spray. Do not get in eyes, on skin, or on clothing. Do not ingest. If swallowed then seek immediate medical assistance.

**Storage.** Keep containers tightly closed in a dry, cool and well-ventilated place. Corrosives area. Incompatible Materials. Metals. Acids. halocarbons.

## 8. Exposure controls / personal protection

### Exposure Guidelines

| Component        | Alberta                      | British Columbia             | Ontario TWA/EV           | Quebec                       | ACGIH TLV                    | OSHA PEL                                                 | NIOSH                                                      |
|------------------|------------------------------|------------------------------|--------------------------|------------------------------|------------------------------|----------------------------------------------------------|------------------------------------------------------------|
| Sodium hydroxide | Ceiling: 2 mg/m <sup>3</sup> | Ceiling: 2 mg/m <sup>3</sup> | CEV: 2 mg/m <sup>3</sup> | Ceiling: 2 mg/m <sup>3</sup> | Ceiling: 2 mg/m <sup>3</sup> | Ceiling: 2 mg/m <sup>3</sup><br>TWA: 2 mg/m <sup>3</sup> | IDLH: 10 mg/m <sup>3</sup><br>Ceiling: 2 mg/m <sup>3</sup> |

### Legend

ACGIH - American Conference of Governmental Industrial Hygienists

OSHA - Occupational Safety and Health Administration

NIOSH: NIOSH - National Institute for Occupational Safety and Health

### Engineering Measures

Use only under a chemical fume hood. Ensure that eyewash stations and safety showers are close to the workstation location.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

#### Eye Protection

Goggles

#### Hand Protection

Protective gloves

| Glove material | Breakthrough time | Glove thickness | Glove comments                                         |
|----------------|-------------------|-----------------|--------------------------------------------------------|
| Neoprene       | > 480 minutes     | 0.45 mm         | As tested under EN374-3                                |
| Butyl rubber   | > 480 minutes     | 0.35 mm         | Determination of Resistance to Permeation by Chemicals |

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

### **Respiratory Protection**

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Recommended Filter type:** Particulates filter conforming to EN 143

When RPE is used a face piece Fit Test should be conducted

### Environmental exposure controls

Prevent product from entering drains.

### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice.

## 9. Physical and chemical properties

|                                        |                          |
|----------------------------------------|--------------------------|
| Physical State                         | Liquid                   |
| Appearance                             | Clear                    |
| Odor                                   | Odorless                 |
| Odor Threshold                         | No information available |
| pH                                     | 14 @ 20°C Alkaline       |
| Melting Point/Range                    | < 0 °C / 32 °F           |
| Boiling Point/Range                    | approx 120 °C / 248 °F   |
| Flash Point                            | Not applicable           |
| Evaporation Rate                       | No information available |
| Flammability (solid,gas)               | Not applicable           |
| Flammability or explosive limits       |                          |
| Upper                                  | No data available        |
| Lower                                  | No data available        |
| Vapor Pressure                         | 14 mmHg                  |
| Vapor Density                          | > 1.0                    |
| Specific Gravity                       | 1.182                    |
| Solubility                             | Soluble in water         |
| Partition coefficient; n-octanol/water | No data available        |
| Autoignition Temperature               | No information available |
| Decomposition Temperature              | No information available |

## Viscosity

No information available

## 10. Stability and reactivity

|                                         |                                                                                                              |
|-----------------------------------------|--------------------------------------------------------------------------------------------------------------|
| <b>Reactive Hazard</b>                  | None known, based on information available                                                                   |
| <b>Stability</b>                        | Stable under normal conditions.                                                                              |
| <b>Conditions to Avoid</b>              | Incompatible products. Excess heat.                                                                          |
| <b>Incompatible Materials</b>           | Metals, Acids, halocarbons                                                                                   |
| <b>Hazardous Decomposition Products</b> | Thermal decomposition can lead to release of irritating gases and vapors                                     |
| <b>Hazardous Polymerization</b>         | Hazardous polymerization does not occur.                                                                     |
| <b>Hazardous Reactions</b>              | None under normal processing. Contact with metals may evolve flammable hydrogen gas.<br>Corrosive to metals. |

## 11. Toxicological information

Acute Toxicity**Product Information****Oral LD50**

Based on ATE data, the classification criteria are not met. ATE &gt; 2000 mg/kg.

**Dermal LD50**

Based on ATE data, the classification criteria are not met. ATE &gt; 2000 mg/kg.

**Mist LC50**

Based on ATE data, the classification criteria are not met. ATE &gt; 5 mg/l.

**Component Information**

| Component        | LD50 Oral               | LD50 Dermal           | LC50 Inhalation |
|------------------|-------------------------|-----------------------|-----------------|
| Water            | -                       | -                     | -               |
| Sodium hydroxide | 140 - 340 mg/kg ( Rat ) | 1350 mg/kg ( Rabbit ) | Not listed      |

**Toxicologically Synergistic Products** No information available

Delayed and immediate effects as well as chronic effects from short and long-term exposure

**Irritation** Causes burns by all exposure routes

**Sensitization** No information available

**Carcinogenicity** The table below indicates whether each agency has listed any ingredient as a carcinogen.

| Component        | CAS-No    | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|------------------|-----------|------------|------------|------------|------------|------------|
| Water            | 7732-18-5 | Not listed | Not listed | Not listed | Not listed | Not listed |
| Sodium hydroxide | 1310-73-2 | Not listed | Not listed | Not listed | Not listed | Not listed |

**Mutagenic Effects** No information available

**Reproductive Effects** No information available.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** Respiratory system

**STOT - repeated exposure** None known

**Aspiration hazard** No information available

**Symptoms / effects, both acute and delayed** Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** The toxicological properties have not been fully investigated.

## 12. Ecological information

### Ecotoxicity

Contains no substances known to be hazardous to the environment or that are not degradable in waste water treatment plants. Large amounts will affect pH and harm aquatic organisms.

| Component        | Freshwater Algae | Freshwater Fish                                     | Microtox   | Water Flea |
|------------------|------------------|-----------------------------------------------------|------------|------------|
| Sodium hydroxide | Not listed       | LC50: = 45.4 mg/L, 96h static (Oncorhynchus mykiss) | Not listed | Not listed |

**Persistence and Degradability** Soluble in water Persistence is unlikely based on information available.

**Bioaccumulation/ Accumulation** No information available.

**Mobility** Will likely be mobile in the environment due to its water solubility.

## 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

## 14. Transport information

### DOT

UN-No UN1824  
 Proper Shipping Name SODIUM HYDROXIDE SOLUTION  
 Hazard Class 8  
 Packing Group II

### TDG

UN-No UN1824  
 Proper Shipping Name SODIUM HYDROXIDE SOLUTION  
 Hazard Class 8  
 Packing Group II

### IATA

UN-No UN1824  
 Proper Shipping Name SODIUM HYDROXIDE SOLUTION  
 Hazard Class 8  
 Packing Group II

### IMDG/IMO

UN-No UN1824  
 Proper Shipping Name SODIUM HYDROXIDE SOLUTION  
 Hazard Class 8  
 Packing Group II

## 15. Regulatory information

### International Inventories

| Component        | CAS-No    | DSL | NDSL | TSCA | TSCA Inventory notification - Active-Inactive | EINECS    | ELINCS | NLP |
|------------------|-----------|-----|------|------|-----------------------------------------------|-----------|--------|-----|
| Water            | 7732-18-5 | X   | -    | X    | ACTIVE                                        | 231-791-2 | -      | -   |
| Sodium hydroxide | 1310-73-2 | X   | -    | X    | ACTIVE                                        | 215-185-5 | -      | -   |

| Component | CAS-No    | IECSC | KECL     | ENCS | ISHL | TCSI | AICS | NZIoC | PICCS |
|-----------|-----------|-------|----------|------|------|------|------|-------|-------|
| Water     | 7732-18-5 | X     | KE-35400 | X    | -    | X    | X    | X     | X     |

|                  |           |   |          |   |   |   |   |   |   |
|------------------|-----------|---|----------|---|---|---|---|---|---|
| Sodium hydroxide | 1310-73-2 | X | KE-31487 | X | X | X | X | X | X |
|------------------|-----------|---|----------|---|---|---|---|---|---|

**Legend:**

X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

DSL/NDL - Canadian Domestic Substances List/Non-Domestic Substances List

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

**Canada**

SDS in compliance with provisions of information as set out in Canadian Standard - Part 4, Schedule 1 and 2 of the Hazardous Products Regulations (HPR) and meets the requirements of the HPR (Paragraph 13(1)(a) of the Hazardous Products Act (HPA)).

**Other International Regulations****Authorisation/Restrictions according to EU REACH**

| Component        | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|------------------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Sodium hydroxide | -                                                                   | Use restricted. See item 75. (see link for restriction details)               | -                                                                                                     |

**REACH links**
<https://echa.europa.eu/substances-restricted-under-reach>
**Safety, health and environmental regulations/legislation specific for the substance or mixture**

| Component        | CAS-No    | OECD HPV | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|------------------|-----------|----------|------------------------------|---------------------------|--------------------------------------------|
| Water            | 7732-18-5 | Listed   | Not applicable               | Not applicable            | Not applicable                             |
| Sodium hydroxide | 1310-73-2 | Listed   | Not applicable               | Not applicable            | Not applicable                             |

| Component        | CAS-No    | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|------------------|-----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Water            | 7732-18-5 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |
| Sodium hydroxide | 1310-73-2 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Annex I - Y35                      |

## 16. Other information

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**Revision Summary**

This document has been updated to comply with the requirements of WHMIS 2015 to align with the Globally Harmonised System (GHS) for the Classification and Labelling of Chemicals.

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**